

General Instructions:

- This question paper consists of 6 sections with a total of 24 questions.
- You are required to fill your basic details - **roll number, email id and phone number** in **section 1** with utmost care.
- The **marking scheme** is given in **Section 1**, one of the unique schemes you might ever experience.
- No ambiguity related to the marking scheme will be entertained later.
- If any discrepancy in the paper is found, the final decision regarding the same will be taken by Team Technothlon.
- Integer type answers can have any integer value (NOT restricted to 0 to 9)
- The Marking scheme given for the particular question on the screen shall **not** be considered for evaluation and the marking scheme given in **section 1** shall be considered for evaluation.

Example: the marking scheme of the question on screen is +4/0 but for evaluation we will consider the marking scheme that is mentioned in **Section 1**.

- Please ignore the auto-generated leaderboard/marks (If any) . You will get the prelims results on the Technothlon home page [https:// technothlon.techniche.org.in/](https://technothlon.techniche.org.in/), where you registered for the test.

Selection Criterion and Result:

- The ranking will be based on the total marks obtained in all the sections.
- The result will be declared on or before August 8th, 2023 on our website technothlon.techniche.org.in.
- To check your result, log in with the Roll number and password provided via email.
- The top 50 participants from your squad will be invited to IIT Guwahati for the Mains and will be awarded Gold certificates.
- The next 200 will be awarded Silver certificates.

Section 1

QUESTION 1

"Legend tells of a legendary warrior whose kung fu skills were the stuff of legend..." A mysterious panda donning a flowing cape and sedge hat walks through the Chinese landscape. Reputed to be a grand master of kung fu, he enters a tavern where he is immediately attacked by the local ruffians. However, they are no match for his skill and even the legendary kung fu masters, the Furious Five, bow down to the panda's skill, requesting to hang out with him and fight alongside him as...

...Po the giant panda wakes up from his dream in his room. His goose father, Mr. Ping calls to him from the noodle restaurant below to help serve tables. Po admires his Furious Five action figures before going downstairs. An extreme fan of Kung Fu, Po dreams of one day becoming a master worthy of fighting alongside the Five but his girth and

clumsiness makes this dream simply a dream, and his kung fu talents reside only within his knowledge of moves and artifacts. He is hesitant to express his desires to his dad who is more interested in running his restaurant and advertising his famous 'secret ingredient soup'.

Main Character

Po is the main protagonist of the Kung Fu Panda franchise. A kung fu fan and giant panda working at his adoptive father's noodle restaurant.

Side Characters

Master Shifu is short a red panda, the headmaster of the Jade Palace, and the trainer of the Furious Five. He joined the palace when his father, the con artist Shirong, abandoned him there.

Grand Master Oogway is an elderly Galápagos Tortoise, the senior headmaster of the Jade Palace before Shifu, and the creator of kung fu. Highly venerated for his wisdom, knowledge, and experience.

Master Tigress is a South China tiger and the leader of the Furious Five. She was adopted by Shifu who disapproved of her aggression and strength, wanting her to emphasize his control and precision. Oogway, on the contrary, encouraged Tigress to grow into her own.

Master Viper is a green tree viper with two small lotus flowers on top of her head. Although she was born with ineffective venomous fangs, she compensates with her strength and precision.

Master Monkey is a golden snub-nosed monkey with a thick Chinese accent, and the most humorous of the Five. Unlike the other members of the Five, Monkey prefers to use a weapon in combat.

Master Mantis is a Chinese mantis and the smallest of the Five. He is, however, the strongest proportional to his size, and can move at superhuman speeds.

Master Crane is a black-necked crane and the most patient of the Five. He serves as a scout and lookout in combat situations, as well as a means of transport for the other members of the Five.

Mr. Ping is a Chinese goose and the owner of the most popular noodle shop in the Valley of Peace. He adopted Po when he found a baby panda in one of the shop's radish crate deliveries.

Zeng is a swan goose and a messenger of Master Shifu. He is given the mission to double the amount of guards at Chorgh-Gom Prison and prevent Tai Lung's escape.

Li Shan is a giant panda and the biological father of Po. He was separated from his son when Lord Shen's wolf army attacked their village and was presumed dead with the other pandas.

Villain

Tai Lung is the main antagonist of Kung Fu Panda. He is a muscular snow leopard with supernatural strength. Shifu's adopted son, Tai Lung was sent to Chorh-Gom Prison for twenty years as a result of his crimes by turning to the dark side.

Marking Schemes

Power Sequence (This marking scheme is applied to section 2)

Power Scheme: With negative marking

For consecutive correct questions, you will be awarded 21, 22, 23 and so on.

However, if you leave a question or answer it incorrectly, the sequence is broken, and you will start again from 21.

If you answer the question incorrectly your marks will be deducted in the same pattern but different sequence

E.g. Solving 4 questions,

For all correct you get $2+4+8+16 = 30$

For all incorrect you get $-1-2-4-8 = -15$

For RRWW you get $2+4-1-2 = 3$

For RWRW you get $2-1+2-1 = 2$

For WRWW you get $-1+2-1-2 = -2$

For RRUW you get $2+4+0-1 = 5$

And so on.

(R -> right answer, W-> wrong answer, U-> Unattempted)

Affinity In A Row (This marking scheme is applied to section 3)

Q1, Q2, Q3 have marks 2, 4, 6 and so on respectively BUT

There are two conditions:

There would be no reduction in the marks of this section if you solve 2 consecutive questions correctly

Also, If the above criteria is not met alongside you solving a question incorrectly, the total marks you have scored in this section are halved.

For example :

For a 3 question section-

If you solve Q2 and Q3 correctly but Q1 incorrectly or leave it unattempted

then you will get $0 + 4 + 6 = 10$ marks.

If you solve Q1 and Q3 correctly but Q2 incorrectly or you have not attempted it, then you will get $(2 + 0 + 6)/2 = 4$ marks.

Similar rules are applied for a section containing 4 questions with the last question having 8 marks.

Similar rules are applied for a section containing 5 questions with the last question having 10 marks.

Shield Me (This marking scheme is applied to section 4)

You get 5 marks for answering correctly and 4 marks are deducted for answering incorrectly, but if you have solved the previous question correctly, then there is no negative marking for the current question.

For example:

If Your sequence is RWRW, then your score is $5+0+5+0 = 10$

But if your sequence is RWWR, then your score would be $5+0-4+5 = 6$

And if your sequence is RWUW, then your score is $5+0+0-4 = 1$

Where (R -> right answer, W-> wrong answer, U-> Unattempted)

Arithmetic Game (This marking scheme is applied to section 5)

You will be awarded 2,4,6,8 for consecutive correct answers and a negative of

-1,-2,-3,-4 for incorrect answers.

For example:

If Your sequence is RRRW, then your score is $2+4+6-4 = 8$

But if your sequence is RWRR, then your score would be $2-2+6+8 = 14$

And if your sequence is WRUR, then your score is $-1+4+0+8 = 11$

Where (R -> right answer, W-> wrong answer, U-> Unattempted)

Bonus Questions

There are some bonus questions to boost your score in the competition.

The Bonus question is applicable only if the previous question is correct and the marks of the bonus question will be the same as the score you obtained in the previous question.

There is no negative marking for the Bonus Questions.

Note: Bonus Questions are not included in the marking scheme.

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It is assumed that you have read the instructions properly and Best Of Luck!!

Enter Your Technothon'23 Roll Number. Enter the digits after HE..... or JE.....

(Roll Number was shared via your registered email address. Kindly check the Spam Folder)

(Fill it very Carefully)

Answer.....

QUESTION 2

Enter Your Technothon'23 registered Phone Number (10 Digits) .

(Phone number on which you received the OTP)

(Fill it very Carefully)

Answer.....

Section 2

QUESTION 3

Po goes on to serve the tables in his father's restaurant, and finds a rather peculiar sight. Po's father is greatly alarmed by the wolf's menacing promise to destroy the restaurant and begins pleading with Po to come up with a solution. Po presents him with a puzzle because he thinks that the wolf can only be destroyed by knowledge, not via a physical conflict. The puzzle follows:

Po then goes on to explain that a word consisting of only lowercase letters is called alphabetical if deleting or inserting zero or more of its letters can result in the alphabet string "abcdefghijklmnopqrstuvwxyz". Given a word, determine the minimum number of letters to insert or delete anywhere in the string to make it alphabetical. Deletions do not increase the count. The given word is "axstcgdsuv".

- B. 20
- C. 21
- D. 18

QUESTION 4

Po's quick thinking and wits help him as the wolf is quite annoyed with the difficult question and decides to leave the restaurant without any violence. Po's dad and the other animals at the place applaud him for his intelligence.

Meanwhile, Master Shifu, a red panda who resides at the Jade Palace temple, practices kung fu in the courtyard with his students, the Furious Five; Tigress, Crane, Monkey, Viper, and Mantis, before he is summoned to see tortoise Grand Master Oogway. Fearing something is wrong, Shifu rushes to find him meditating in the grand hall. Oogway doesn't deny bad news but calmly tells Shifu he's predicted that the snow leopard warrior Tai Lung will escape prison and return to the Valley of Peace to wreak destruction once again. Intent on never letting this happen, Shifu sends his goose messenger, Zeng, to fly to Chor-Gom Prison to ensure that security is increased. Oogway then proclaims that it is time to choose the Dragon Warrior; a master of great skill who will be granted the secrets of the universe by reading the Dragon Scroll kept delicately out of reach on the temple ceiling. Assuming that one of the Furious Five will be chosen, Shifu prepares a competition to determine which one will qualify.

Flyers for the competition are spread throughout town and the villagers flock to the temple. The news eventually reaches Po and he is excited to have the opportunity to see his idols in person, but before that, he starts to dress properly for the occasion, He wears nice new shorts and struggles to decide what kind of socks to wear and is met with a dilemma

Po has 98 different types of socks in his wardrobe. Po then remembers there are **i** socks (not pairs) of the **ith** type. For example, there are 46 socks of the 46th type of socks. How many maximal **mismatched** pairs can he get from his wardrobe?

- A. 2420
- B. 2415
- C. 2410
- D. 2425

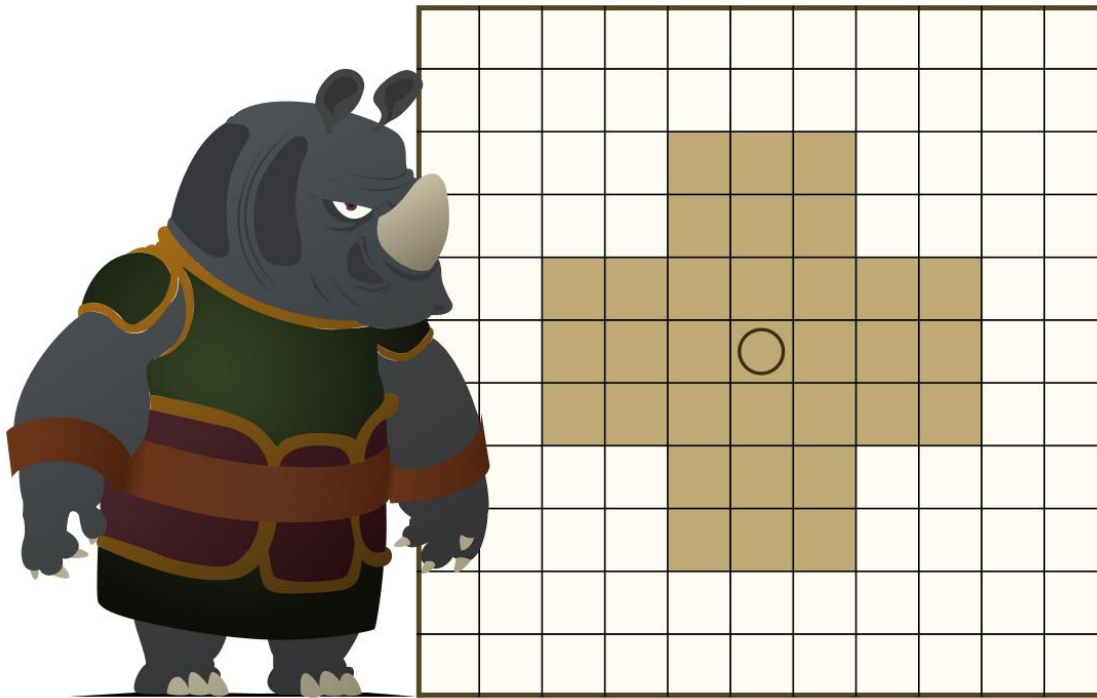
QUESTION 5

Eventually, he settles on his favorite pair and starts his journey toward the temple, Po follows with his noodle cart in tow but struggles on the long staircase to the temple. He is the last to arrive at the gates and finds that they've shut him out.

He then approaches the guard rhino right outside the gate and asks him to help. The rhino is quite annoyed at the fact that a fat panda wants to go see a martial arts and kung fu tournament so he proceeds to trick Po and says he'll get the entry only if he solves a riddle. With no other choice left, Po accepts the rhino's challenge and is presented with the following riddle

The rhino explains his hypothetical scenario consisting of an infinite grid of squares.

In one of the squares, there lives a cell (Biological unit of life) (marked as a circle in the following pictures).



Cells cannot move, but they can perform their unique action: a cell can split itself into two daughter cells, which are identical to the original one, and each will occupy a square that is (orthogonally) adjacent to the original square.

Since every square can only accommodate only one cell, a splitting can only happen when the cell has at least two empty adjacent squares (if there are more than two, then it can choose freely which squares to split). Also, two cells should not split simultaneously, so that no conflict should occur.

On the grid, there is a region called "the prison" (painted grey in the following pictures). The aim is to let the cells escape the prison, i.e. to reach a status that no cell is in the prison.

The guard then asks Po to answer the minimum number of daughter cells that will be there at the end when all the cells are free from prison.

- A. 29
- B. 41
- C. 38
- D. 44

QUESTION 6

Po is perplexed by the complicated question, so frees his mind and calmly starts thinking of a solution but as you'd expect from Po, he fumbles the wrong answer out, even though he is quite confident about the solution he chose.

The guard goes on with his duty and denies him entry inside. Following that in a desperate bid to see the competition before it's over, Po fashions fireworks to his cart and blasts himself high into the air, only to come crashing down in the arena just as Oogway is preparing to choose out of the Five.

Po has unwittingly gatecrashed the tournament for identifying the Dragon Warrior. However, instead of expelling him from the arena, Master Oogway decides that Po is his successor, pointing his finger toward him and announcing him as the next Dragon Warrior, much to the shock of Master Shifu and the Furious Five and the entire audience. Shifu believes that Oogway's choice is an accident, and advises him to reconsider his decision.

Oogway proclaims that a true kung-fu warrior must not just be powerful in his physical body, but should also be mentally strong, and guess his opponents' moves. He proposes a card game – which has a historical importance in China – of a new variant called 20-card. The game will be played one by one, by each of the Furious Five, along with Po. The player must either select a card or pass it on at every turn, with the goal of maximizing the sum of the numbers on the chosen card. However, there is one rule to keep in mind: **if four cards are chosen in consecutive turns, the next two cards must be passed on.**

The players are told the order in which the numbers on the cards will appear:

5, 3, 5, 4, 8, 7, 6, 6, 10, 15, 13, 14, 15, 5, 3, 6, 6, 8, 9, 7

(To illustrate the rule more clearly, if Po selects 5, 3, 5, 4, he has to pass on 8 and 7 – the next two cards).

Each of the Furious Five take turns, with Master Tigress achieving the highest score among them. However, Po believes he can better her to reach the maximum score. What score does Po reach?

- A. 125
- B. 141
- C. 97
- D. 107

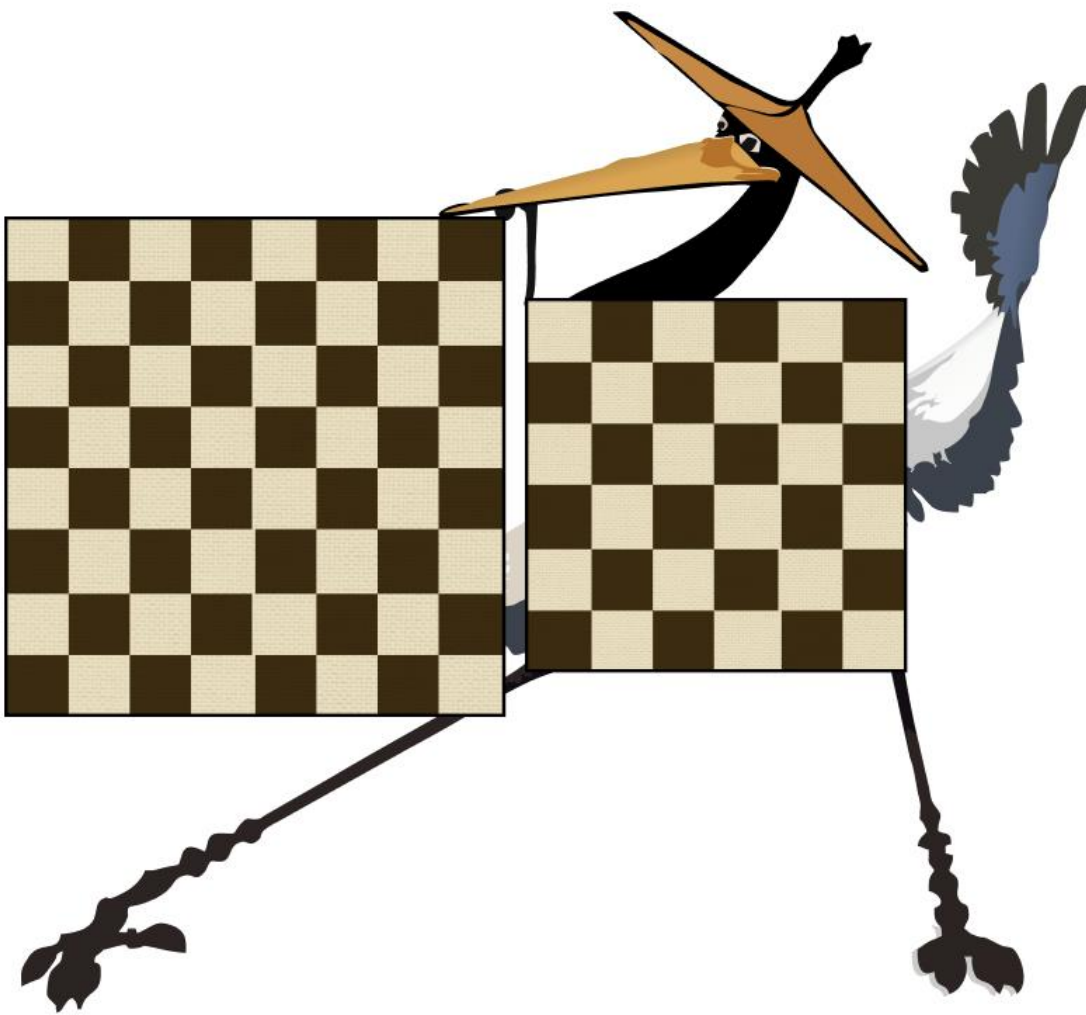
Section 3

QUESTION 7

To everyone's utter surprise Po achieves a higher score than Tigress and is announced by Master Oogway as the next Dragon Warrior. Oogway's decision is final, despite Shifu's protests, and Po is carried (unsuccessfully) into the temple leaving Tigress greatly disappointed since it seemed Oogway would have chosen her if Po hadn't arrived.

Inside the temple, Po examines, with awe, the many weaponry and valuable artifacts before Shifu approaches him and berates him for his obvious lack of skill. He deflates Po's excitement by grabbing his finger in what Po recognizes as the Wuxi Finger Hold, reputed to be extremely powerful. Shifu promises that Po will regret ever being chosen before taking him to the Fives training room where Po is promptly put to the test.

During training each of the furious five wear a certain uniform made of cloth with squares. They come in 2 sizes, the one with 64 squares and the other with 36 squares. Po who is new to the training grounds, finds it really hard to fit in the uniform due to his large attire. Crane suggests him to combine the 2 cloths, one having 64 squares and the other having 36 squares to form a cloth having 100 squares in a 10*10 manner. Po likes the idea to combine the clothes and believes that the resulting cloth would suit him perfectly. Now he has to divide both the clothes into a minimum number of pieces. For his help, Crane tells him the minimum number of pieces would come out to be 4. Now What would be the number of black squares in each of the pieces?



- A. 4,14,16,16
- B. 3,7,14,16
- C. 4,6,14,26
- D. 5,15,4,26

QUESTION 8

Eventually the newly made cloth suits Po perfectly as he nervously but excitedly goes on to try some kung fu moves, Po accepts the challenge but he is hurtled, flung, and beaten down to a now-existing Level 0. On the way to the dormitories, Po overhears the Five poke fun at his incompetence. He has an awkward conversation with Crane, opening up to his own doubts of being the Dragon Warrior. Tigress assures his doubts, calling him a disgrace to kung fu, and tells him that, if he is wise, he will leave by morning.

Disheartened, Po retreats to a secluded spot. Oogway finds him sitting beneath the Peach Tree of Heavenly Wisdom with peaches stuffed into his mouth. Sensing his distress, and noting that he finds comfort in food, Oogway tells Po

that he worries too much about what was or what could be. He should instead focus on the present moment rather than doubting himself.

To lift his spirit up Oogway tells him about the magic realm. He swirls his staff high in the sky and showcases a gateway to the magic realm. Po is awestruck by what he sees and proceeds to go inside but finds the gate to be locked. He looks at Oogway who smiles at him and replies, to see the greatest of places he must overcome the toughest of tasks. Po walks back in disappointment but Oogway stops him and reminds him to believe in himself and go ahead. He explains the gate has a lock that unlocks if you input the correct four digits, in any order. The lock has a keypad with the digits 0,1,2,3,4,5,6,7,8,9.

For example, if the unlock code is 1000 then 1000, 0100, 0010, 0001 all these codes will open the lock.

How many different unlock codes can be there for Po to successfully open the gateway to the magic realm? (Note: Two unlock codes are equivalent if they contain the same digits. For example 1000 and 0100 are equivalent and counted as one)

- a. 805
- b. 750
- c. 620
- d. 715

QUESTION 9

Po opens the gate successfully and is met with inner peace for a while. He thanks Oogway for inspiring him and helping him while he is down on luck. Oogway goes on to appreciate him and once again reminds him to focus on the present moment rather than doubting himself. Renewed with a sense of determination, Po rises the next morning before anyone else and announces that he will stay to train. He spars with each of the Furious Five, receiving swift beatings despite his undeterred enthusiasm. Frustrated by Po's perseverance, Shifu is at a loss. However, the Five become impressed by Po's fortitude and eventually start to warm up to him, with the exception of Tigress.

Soon the 6 of them take a break for lunch. During which Monkey and Mantis decide to play a fun game in the meanwhile of Master Shifu's absence. There are two piles of dumplings, one containing 101 dumplings while the other contains 201. Essentially they play a game in which they have to pick up dumplings from the piles according to the following rule they came up with:

Remove the number of dumplings from a pile that is equal to one of the divisors of the number of dumplings in the other pile.

The Player removing the last dumpling from the pile wins. Monkey takes up the first pile while Mantis the other and assuming that both of them play optimally (playing to win)

Who will win? (Answer 1 for Monkey and 2 for Mantis) **(Integer Type)**

Answer.....

QUESTION 10

Bonus Question (only applicable if the answer to the previous question is correct)

Who will win if the number of dumplings in the first pile is decreased by 1? (100 and 201 dumplings) **(Integer Type)**

Answer.....

QUESTION 11

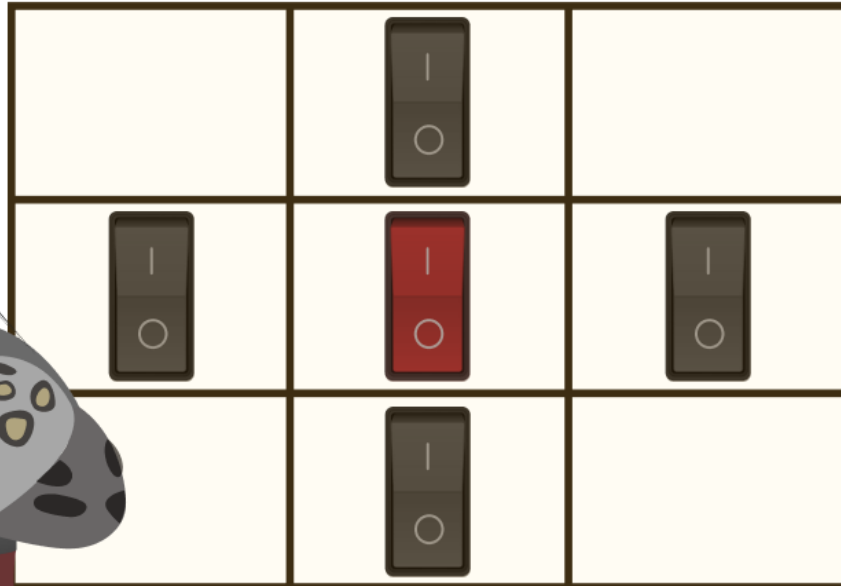
Following that, the Five reveal that Shifu was not always as stern as he currently is. Tigress explains that Tai Lung himself was the reason for Shifu's shift in personality. Having found him as a newborn on the steps of the temple, Shifu raised Tai Lung as his own and began to hope that he was the fabled Dragon Warrior when the cub displayed a natural talent in kung fu. However, when the time came to stand before Oogway, the grand master sensed the darkness in Tai Lung and denied him the honor of receiving the Dragon Scroll. Enraged, Tai Lung went on a rampage through the village and attempted to take the scroll by force. Caring for him too much, Shifu was unable to fight Tai Lung and Oogway was forced to defeat him by blocking his chi. Tigress then says that Shifu never cared for anything or anyone since, not even her; his own foster daughter.

Meanwhile, Zeng arrives at Chor-Gom Prison and is escorted to Tai Lung's hold by the over-confident Commander Vachir, a rhinoceros. Vachir assures Zeng that the prison is inescapable, and explains that even if someone has to break free of everything, they'll be met with a series of three traps and puzzles which are deemed to be unsolved.

But luck, as it were one of Zeng's feathers, falls within reach of Tai Lung. Using his tail, he uses the feather to remove the chi-blockers in the turtle-shell contraption on his back and break free.

Shocked by that, Commander Vachir orders his guards to lock all entrances and start an attack on Tai Lung. Meanwhile, Tai Lung shows an explosive display of his Fierce Kung Fu abilities and thrashes anyone who comes in his way. Soon he reaches the main entrance where he is tested with a series of puzzles

Tai Lung first finds himself with a switchboard which is in the shape of a square, 6 dm (decimeter) in length. Each 1 dm² area is occupied by a switch. Every switch is in an off state initially. He quickly deduces that he must turn on the maximum number of switches possible and goes on to toggle one of the switches and notices that if you toggle a switch then along with it the switches which are orthogonally adjacent to it also get toggled.



Following the above rule, how many maximum numbers of switches can Tai Lung switch on?

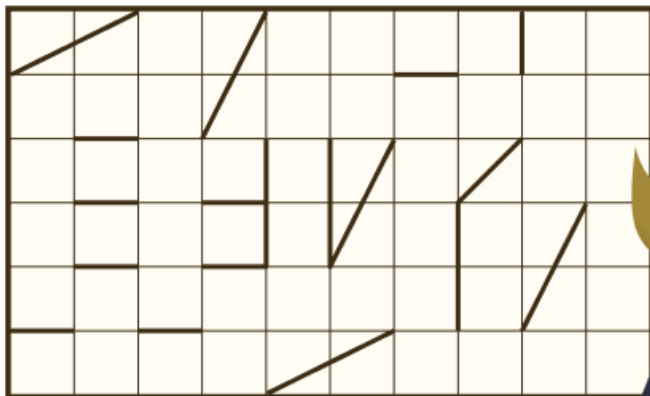
Toggle If a switch is in the off state it becomes in the On state and vice-versa.

Orthogonal: If we toggle the switch at the center (red) then the switches in black will also get toggled. **(Integer Type)**

Answer.....

QUESTION 12

Tai Lung showcases his magnificent problem solving skills as he turns on the required number of switches which open the door ahead leading him to the second puzzle. A lot of guards and commander Vachir soon surround him, whom he defeats easily and threatens Vachir to help him solve the next puzzle. Vachir, helpless as he were goes on to explain the puzzle.



A grid as shown above lies in front of them. The commander instructs Tai Lung to enter the exact number of shaded cells according to the 2 rules stated

1. Two shaded cells cannot share an edge, and all unshaded cells must be orthogonally connected.
2. Once finished, it is not possible to draw a line segment in the grid which passes through two borders (bolded lines) without passing through a shaded cell.

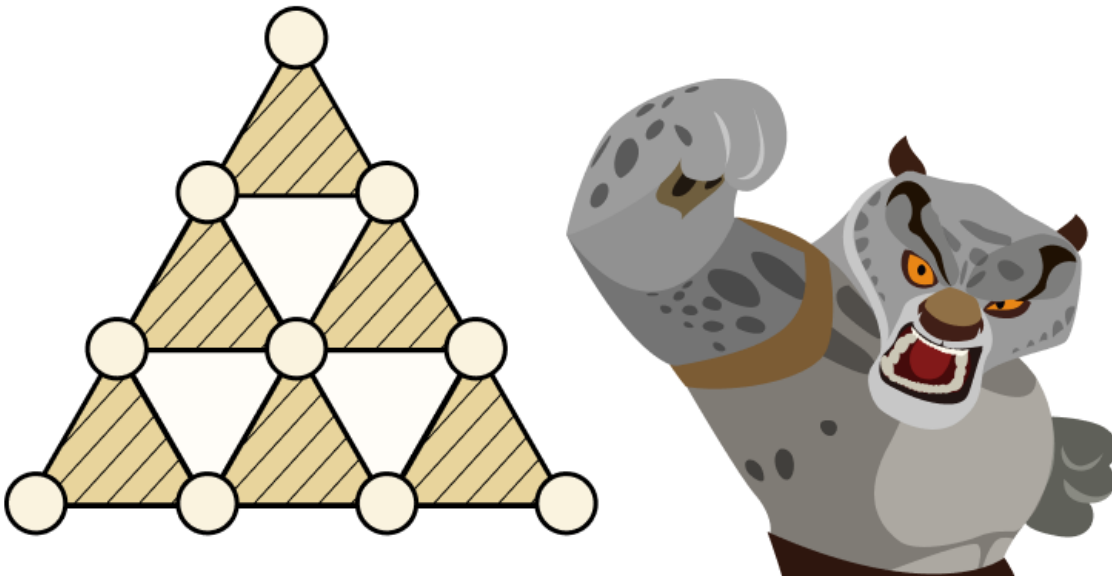
He quickly solves it as the rules stated. So What was the number he entered? **(Integer Type)**

Answer.....

Section 4

QUESTION 13

Tai Lung is eventually faced with the final challenge before he escapes the prison. He grabs Vachir by neck as he reveals that in order to open the final door, it is required to fill numbers from 0 to 9 exactly once in each of the above circles such that the sum of numbers at the vertices of the shaded triangles is equal. Find the minimum sum. If not possible answer as 0



Soon Tai Lung is out of the prison and he goes on to grab Zeng, instructing him to fly back to Shifu to announce that the real Dragon Warrior is coming home.

Back at the temple, Po entertains the Five with some noodle soup he prepared. Outside, Shifu is met by a disheveled Zeng who relays the news. Panicked, Shifu finds Oogway beside the Sacred Peach Tree and tells him that Tai Lung is on his way. Oogway is shocked, but assures Shifu that, if he finds faith in the panda and trains him, Tai Lung will be defeated and Shifu will finally find inner peace. He then gives Shifu his wooden staff and claims that his time has come. He walks near the edge of the cliffside and vanishes in a haze of peach blossoms.

Shifu then informs Po and the Five of Tai Lung's escape and tells Po he is the only one who can stop him. Feeling cheated, Tigress angrily attempts to explain to Shifu that he has been training the Five for many years to prepare for Tai Lung's escape, but Shifu tells her it is Po's destiny to stop Tai Lung. Upon hearing that Oogway has left and that Tai Lung is approaching quickly, Po, still unable to make any progress with kung fu, attempts to run away, but Shifu stops

him. Po then makes Shifu admit that he does not know how to train the panda into the Dragon Warrior. Tigress overhears this and leads the Five in a secret attempt to stop Tai Lung themselves. **(Integer Type)**

Answer.....

QUESTION 14

Meanwhile, Shifu discovers that Po is capable of impressive physical feats when motivated by food, and decides to train Po by incorporating food to develop a unique method of imparting his kung-fu knowledge to Po. However, Shifu explains to Po that Master Oogway had taught him certain fundamentals before he started his kung fu journey and for Po to become the Dragon warrior, he needs to learn it too.

The first fundamental being the 'Enchanted IO Language'. This language will be incorporated in MahJong Tiles which Po has to be proficient in. Shifu explains that there are only 2 letters in the alphabet of the IO language I and O and for Po to understand better, he substitutes I for the riceball and O for dumpling. Shifu says any dish will consist of combinations of riceball or dumplings lined together. If he removes two neighbouring components, "riceball and dumpling" from any plate then he will get a dish with the same meaning. Similarly, the dish will not change if he inserts combinations of "dumpling, riceball" or "riceball, riceball, dumpling, dumpling".

Po has been given a word having a combination of **IOOIIIIOIIOOOIOIOIO**, and has to pick tiles having words of the same meaning so as to win the game of MahJong. Help Po to find the number of words from the given list that have the same meaning as that of **IOOIIIIOIIOOOIOIOIO**

1. IIOIIOOOIIOIOOOIOIOI
2. OIOIOIOIOIOIOIOIOIOI
3. OIIIOOOIOOOOIIOIIII
4. OOOIOOOIOIOIOIIIOIII
5. IOIOIOOOIIOIOOOIIOIOIO
6. IIIIOOIOOOIIOIOOOIOOOI

- a. 2
- b. 3
- c. 5
- d. 4

QUESTION 15

Master Shifu is quite impressed by Po as he correctly solves the tricky question, Meanwhile The Five meet Tai Lung head on. They find him crossing the Thread of Hope, a series of bridges through the mountains. They attempt to stop him on the last bridge. After an epic battle, the Five gain the upper hand and appear to have defeated Tai Lung, tying him to the remains of the severed bridge and sending him into the mist below. However, Tai Lung's skills prove superior, and he surprises them by flinging himself onto the cliff and overpowering them with a chi block. Crane

carries the paralyzed Five back to the Jade Palace, where Shifu revives them and decides that Po is ready to receive the Dragon Scroll.

With Tai Lung having defeated the Furious Five, Master Shifu believes that the Valley of Peace needs Master Oogway's prediction of the Dragon Warrior to be true. He decides to guide Po to the Dragon Scroll. The two of them enter a labyrinth with a door on the opposite end, which is supposed to lead to the scroll. The path between the two doorways is divided into a 6-by-6 grid, with some large balls in each of the grid squares. One of the squares was glowing. Without thinking about all this, Po crosses over to open the door and tries to open it; however, the door is sealed and all his efforts are in vain. He notices that the door has a slot in it, just enough to fit any ball.

Suddenly, a whisper fills the room – the voice of Master Oogway. He says the following:

You must choose certain squares of the grid, such that the total number of balls appearing in a square can only occur once. You cannot select a total of more than 9 balls from any row or column. These balls you select are the holy ones and must be inserted into the slot to open the door to the Dragon Scroll. One of the grids is glowing and you must select the balls in that grid.*

*For example – if Po selects a particular grid with 8 balls, he cannot select any other grid with 8 balls.

The grid on the floor is depicted below, with the number in each square depicting the number of balls in that grid, and the marked grid being the glowing grid (which must be selected).

	(1)	(2)	(3)	(4)	(5)	(6)
(1)	6	1	6	8	7	9
(2)	5	5	4	6	1	2
(3)	7	9	3	9	4	1
(4)	6	8	7	4	8	5
(5)	7	9	3	6	3	3
(6)	7	2	1	7	8	2

The number in the bracket represents the row and column number and should not be considered for the selection.

Suppose a cell is represented by the coordinates (X, Y) then find the sum of all X coordinates and the sum of all Y coordinates of the balls which Po needs to select in order to open the door.

If the sum of X coordinates = 28,

the sum of Y coordinates = 35,

Give your answer as 2835.

A. 2527

B. 2428

C. 2529

D. 2530

QUESTION 16

Po solves the preceding puzzle and obtains the legendary Dragon scroll. Shifu instructs him to read it, but Po exclaims in surprise to find nothing but a blank, reflective glaze on the scroll. Terribly disappointed, Shifu relents to have the Five evacuate the village while he remains at the temple to hold Tai Lung off. The Five reluctantly agree and Po returns to his father who is packing up the noodle shop. Ping tries to console Po, but Po admits that he's not sure of who he is and can't believe he's Ping's son. Taken aback, Ping whispers to Po something he should have told him long ago; the secret ingredient to his famous secret ingredient soup. The secret is... nothing. He explains that something becomes special if people truly believe that it is. Po opens the scroll to see his reflection on its glazed surface and realizes the secret before turning back to the temple.

At the Jade Palace, Tai Lung overpowers Shifu and demands to know the location of the Scroll. Po arrives with the Dragon Scroll and challenges Tai Lung to combat. Po proves to be a formidable opponent, frustrating Tai Lung with his confusing fighting techniques. After a protracted battle, Tai Lung finally wins the scroll and triumphantly opens it for the supposed power within. However, upon seeing the golden reflection within, he completely fails to understand its symbolic meaning. When Po rises to explain it, Tai Lung refuses to accept the panda's words or the fact that everything the snow leopard has betrayed and sacrificed in his eyes has been for apparently nothing.

Now consumed with such enraged despair, Tai Lung attacks Po again, but without discipline or real purpose behind it. Using ingenuity and his rebounding stomach, Po manages to render Tai Lung helpless. Tai Lung spews out a slur of insults before Po takes his finger in the Wuxi Finger Hold. Tai Lung calls Po's bluff and Po admits that Shifu never taught him the move - he figured it out on his own. Tai Lung who is quite familiar with the move makes up his mind that he isn't going to win over Po by sheer power anymore and in a cunning fashion challenges Po to defeat him in his intelligence and proceeds with a question

He explains that there are N doors numbered from 1 to N in a row and there are N persons. 1st person goes and toggles the doors which are numbered as the divisors of N . 2nd person goes and toggles the doors which are numbered as the divisors of $N-1$. Similarly, the i -th person goes and toggles all doors numbered as the divisors of $N-i+1$.

Toggle: Opening a closed door and closing an opened door.

Divisors of N : Numbers that leave a remainder 0 when N is divided from that number.

If $N = 83$, After all N persons toggle the doors, How many doors will be left open?

A. 43

B. 57

C. 64

D. 59

QUESTION 17

Bonus Question (Only applicable if answer to the previous question is correct)

If $N = 54$, After all, N persons toggle the doors, How many doors will be left open?

- A. 26
- B. 33
- C. 37
- D. 42

QUESTION 18

Po, who by now is accustomed to solving so many similar problems, answers correctly with ease and confidence, rendering Tai Lung's last resort useless. With a triumphant "Skadoosh!", Po flexes his pinky, resulting in a magnificent wave of raw energy that resonates throughout the valley and is felt by the Five leading the refugees into the hills. Having presumably killed Tai Lung, Po is cheered on by the returning villagers. Tigress and the rest of the Five bow down to Po, calling him master, before Po runs back to the temple to check on Shifu. He finds Shifu lying still, but alive, on the floor of the temple and tells him he defeated Tai Lung. Shifu thanks him and tells Po that he's brought him peace before closing his eyes. Po calls out to him and Shifu reprimands him, saying that he's merely resting. Po lies down beside him and, after a quiet, meditative moment, asks if he wants to get something to eat. Shifu says, Ok. They enjoy a meal of dumplings together at the Peach Tree of Heavenly Wisdom, overlooking a new day while a seed that Shifu had previously planted begins to bloom.

Quite a few months have passed since Po, the Dragon Warrior defeated Tai Lung and Master Shifu announces his retirement from teaching, and names [Po](#) as his successor. Po's first attempts to teach inadvertently injure the Furious Five and Shifu advises Po should try to be more like himself and less like Shifu. During lunch, all six of them sit together whilst playing a familiar game.



They are playing a single player game and now it's time for Po's turn. He is given a 8*8 grid which is divided into 2 halves by a thread. Tigress explains to him that he can place any number of checkers (maximum 32) in any pattern (on any square one on each) below the thread (not above it) before starting the game and the goal is to move one of the checkers above the thread to different levels. Here, level indicates the row number above the thread. The row nearest to the thread is level 1, next row above is level 2 and so on. The checkers can only move by jumping over another checker and in the process the checker across which the jump takes places is removed from the grid. The checkers can jump horizontally and vertically only and not diagonally.

Mantis helps Po by dropping him a hint, that to level 1 you need a minimum of 2 checkers , for level 2 you need a minimum 4 checkers, for level 3 you need a minimum of 8 checkers.

How many minimum number of checkers will be required to reach level 4?

- a. 12
- b. 16
- c. 20
- d. 24

QUESTION 19

Bonus Question (only applicable if answer to the previous question is correct)

How many minimum number of checkers will be required to reach level 5 if the board was 10*10 in size?

- a. 45
- b. 64
- c. 32
- d. Not possible

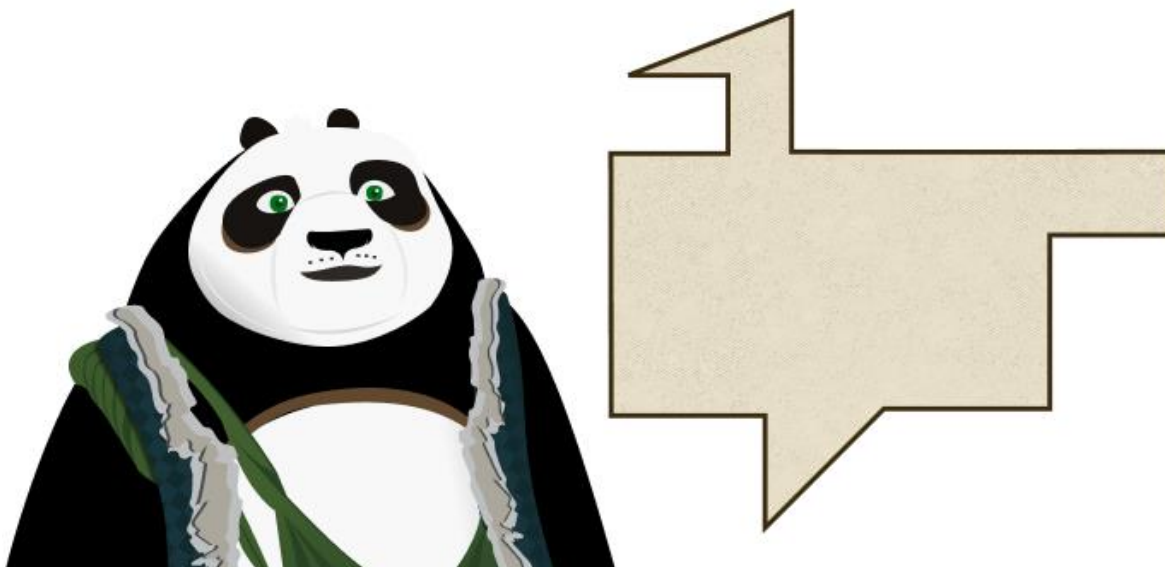
Section 5

Question 20

Meanwhile, the game is interrupted by Crane who tells them that someone is about to beat Po's dumpling-eating record back at his dad's restaurant. Soon Po rushes there to see who might be the record breaker. They go downstairs, but can't see who is eating the dumplings. Once the record is broken, Po sees that it is another panda, bigger even than him. The panda introduces himself as Li Shan, and tells Po that he is looking for his son that he lost many years ago. Po tells him that he lost his father when he was a baby. They wish each other good luck and start to walk away. The other animals in the restaurant shake their heads.

Po and Li stop and look at each other again. They notice each other's green eyes and huge bellies, and realize they're father and son. Li then heartily welcomes Po and his friends to come visit his birth place, The Panda Village. Soon excitedly everyone start their journey out to the hidden panda village. Po and Li, along with the Five are walking through the mountains and decide they should eat lunch. Po finds Mr. Ling inside his lunch sack, and Mr. Ling tells him that he stowed away in case the pandas don't have any food Po likes. They get to the palace, which is behind a giant wall of ice. Li pulls a string, and a bucket comes down. The three of them climb in and are hoisted up into the clouds. Po sees a lush green village with pandas everywhere.

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Answer.....

QUESTION 21

To celebrate his son's return to the panda village Li buys an odd lot of wine in barrels and one barrel containing beer for the party. These are shown in the illustration, marked with the number of gallons that each barrel contained. He sold a quantity of the wine to one Panda and twice the quantity to another, but kept the beer to himself. After a while, he finds it difficult to point out which barrel contains the beer. He decides not to open any to destroy the packaging so he asks Po to find out which one it is, Of course, Li sold the barrels just as he bought them, without manipulating in any way the contents.

(Integer Type)



Answer.....

QUESTION 22

Meanwhile, Po explores the secret panda village revealed to him by his father, Li, and is immediately welcomed by all the other pandas, including many of his relatives that he never knew he had. He spends his time learning Panda habits, like rolling instead of walking, and not doing any physical activity. In celebration of Po's visit, the village organizes a traditional feast with noodles and dumplings. There are many tables with a seating capacity of eight people.

On his table, Po is seated along with his relatives Mei Mei, Bao, Lei Lei, Dim, Sum, Hom-Lee and Meng Meng in a row (not necessarily in the same order). Only 3 among the 8 seated face south while the rest face north. All of them, including Po are of different ages i.e. 49, 28, 36, 84, 16, 18, 77 and 37, but not necessarily in the same order. Four pandas sit between Meng Meng and Dim. Lei Lei sits immediately to the left of Sum and Lei Lei's age is a perfect square as well as an odd number. Dim's age is a perfect square and sits 3rd right of Lei Lei. Bao sits to the right of Lei Lei. Hom-Lee is three times older than Po. Dim's age is double Mei Mei's age. Sum sits 3rd from the extreme end of the row and he is the second oldest person in the group. Bao's age is a perfect square as well as an even number. Lei Lei and Meng Meng both face the same direction. Mei Mei sits 4th left of Lei Lei. Sum faces south. Lei Lei and Meng Meng are immediate neighbors of Hom-Lee. Bao is younger than Dim.

Po asks all his tablemates to talk about themselves, such as their likes, dislikes and ages. Almost no one knows how old they are since life goes on slowly in the village. The 4th person from Bao and the 6th person from Bao are the only ones who know their ages. What is the difference between their ages?

- A. 56
- B. 12
- C. 10
- D. 19

QUESTION 23

Po is understandably taken aback by how many relatives he has and is interested in how many other people have. He then asks Bao the same question. The next response from Bao is rather perplexing. He says that in his family, there are 2 fathers, 2 mothers, 5 offspring, 3 grandsons, 2 sons, 3 sisters, 3 daughters, 1 mother-in-law, 1 father-in-law, and at least 1 niece. How many total members are there in the family?

- A. 23
- B. 15
- C. 8
- D. 7

Section 6

Question 24

Po is now finally happy after all these years, that he met with his family. They enjoy the feast and Po finally finds his motivation to become a teacher and takes on his new role at the restored Jade Palace, where all the pandas and the residents of the Valley of Peace learn kung fu, finally restoring peace and harmony in the valley. He doesn't stop there, Po & his friends are determined to restore harmony all over the world, so they reach their next destination, IIT Guwahati, to spread his teachings of Kung-Fu and inner peace, as many adventures await them during the MAINS EVENT!

On a scale of 1 to 10 how much did you enjoy Po's adventure and how excited are you to continue the adventure at IIT Guwahati at the Mains event?

(10 -> Extremely Enjoyed, 1 -> Did not enjoy at all) **(Integer Type)**

Answer.....